of the rod to the horizontal be . Then

that is,

Substituting the data into gives

At , ball starts sliding outward. Ball starts next to the end, so ball leaves after negligible time. Thus the rod's angle when ball leaves is also

When ball leaves, the massless rod exerts no force on ball . Ball then feels gravity alone. By , its initial speed in projectile motion is

The initial velocity makes the following angle with the horizontal:

Ball stays on the rod during flight; neither exerts a force. The rod follows without affecting it. Ball leaves when its distance from the pivot is again . Take the axes shown. At time , taking the start of ball 's flight as time zero, ball has coordinates

When ball leaves the rod,

Using gives

Solving gives

At this time,